7 **(a)** A parallel beam of light of wavelength 4.5×10^{-7} m and intensity 700 W m^{-2} is incident normally on a surface.

(i) Calculate the number of photons incident per second on an area of 1.0×10^{-4} m^2 of the surface.

number = ...[2]

(ii) Calculate the momentum of a photon of the light.

momentum = ...N s [2]

(iii) Calculate the change in momentum of photons incident on 1.0×10^{-4} m^2 of the surface in one second. Assume that all photons are absorbed by the surface.

change in momentum = ... N s [2]

(b) The orbital electron in a hydrogen atom may be thought of as being trapped in an infinite square potential well such that the electron is confined to a region of width L, as shown in Fig. 7.1.

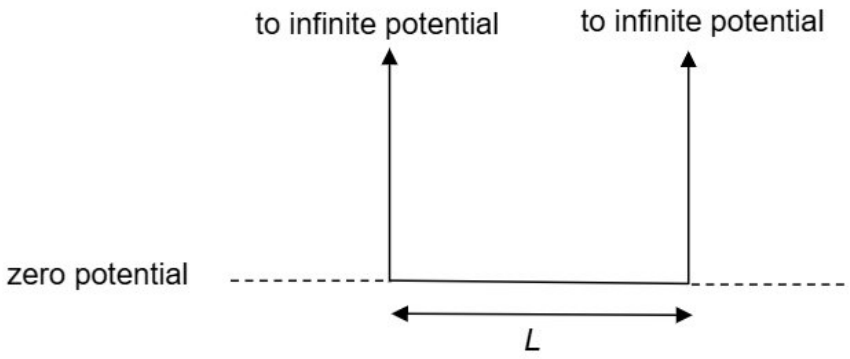


Fig. 7.1

The electron is represented by a standing wave that has wavefunction Ψ given by

$$\Psi = A \sin nkx$$

where x is the distance of the electron from the left-hand edge of the potential well, n is a non-zero integer and A and k are constants that depend on L.

(i) When $n = 1$, the electron is in its ground state.

Determine an expression for k in terms of L.

k = ...[2]

(ii) State what is represented by $|\Psi|^2$.

.....

..... [1]

(iii) Determine an expression for A. Show your working.

[2]

[Total: 11]